

ACTIVITY 8 — FACILITATOR GUIDE

Root Cause Analysis with 5 Whys, Fishbone, and Evidence

TYPE	DURATION	DMAIC PHASE	MAPS TO
Core Lab	60 min	ANALYZE	A3

Purpose of this activity

Identify root causes of variation using structured analysis (A3).

Move from symptom to cause. Apply the 5 Whys to drill past the obvious, organise candidate causes on a Fishbone diagram, then test each candidate against the evidence you actually collected.

What learners produce

A 5 Whys chain ending at an actionable cause, a Fishbone diagram across the five 5M categories, and an evidence-tested cause shortlist.

Materials required

- Learner Worksheet (one per learner)
- Checklist (one per group)
- A laptop with Excel, Google Sheets or LibreOffice Calc
- The CSV files in this activity's data/ and templates/ folders
- Flipchart or A3 sheet and markers
- The course slides for the scenario and the framework
- A browser — this activity uses an online tool (URL in the run-sheet).

Set-up

Groups of 3-5. Distribute one worksheet per learner and one checklist per group. Every activity runs on the same continuous Contoso Service Desk scenario, so remind learners where this one sits in that story before starting.

Scenario / briefing material

Contoso Service Desk handles employee IT requests. Employees complain that tickets take too long to be assigned, that they must keep chasing for status, and that agents apply different definitions of what counts as "resolved". Handoffs between the front line and the specialist queues are unclear, and although ticket data exists it is inconsistent. You are supporting the improvement team as a Yellow Belt — you do not lead the project.

Two weeks of data have been collected: 400 tickets, of which 96 contained at least one defect, with 120 defects recorded in total across 6 defect opportunities per ticket. Mean assignment time is 55

minutes (median 46) against a 30-minute improvement goal. The CSV files in this activity's data/ folder are that extract.

Files in this activity folder

data/ — the mock data you analyse:

- daily-summary.csv — 10 rows (date, tickets, mean_assignment_min, median_assignment_min, defects, missed_30min_goal)
- defect-counts.csv — 6 rows (defect_category, count)
- root-cause-evidence.csv — 8 rows (evidence_id, observation, source, supports_cause)

templates/ — the worksheets you complete:

- cause-evidence-test.csv (candidate_cause, evidence_that_supports_it, evidence_that_contradicts_it, verdict)
- fishbone-causes.csv (category, possible_cause, supported_by_evidence)
- five-whys.csv (why_level, question, answer, evidence_needed, evidence_found)

Open these in Excel, Google Sheets or LibreOffice Calc. Work on a copy so the originals stay clean for revision.

Test every candidate cause against the Lab 7 data before shortlisting it.

How to run it — step by step

#	Instruction to learners	Facilitator note
1	Write the problem statement precisely — the effect you are explaining.	Chains that stop at 'human error' are unfinished — push for the system that allowed it.
2	Complete a 5 Whys chain with the online 5 Whys tool, continuing until you reach an actionable cause. Tool: https://alfredang.github.io/5whys/	The evidence file deliberately contains both supporting AND confounding items.
3	Build a Fishbone diagram with the online Fishbone tool using the 5M categories: Manpower, Method, Machine, Material, Measurement. Tool: https://alfredang.github.io/fishbone/	Do not let the group vote before generating — evaluation kills brainstorming.
4	Brainstorm candidate causes into each category — no evaluation during generation.	
5	Use multi-voting to shortlist the most likely causes as a team.	
6	Test each shortlisted cause against your Lab 7 data — does the evidence support it?	
7	State why the team must not jump straight to solutions.	

Suggested timing

Minutes	Phase
5 min	State the problem precisely
15 min	Complete the 5 Whys chain
15 min	Build the Fishbone across the 5Ms
10 min	Brainstorm causes into each category

8 min	Multi-vote to shortlist
7 min	Test each cause against the evidence

Total: 60 minutes

Discussion & decision prompts

1. Did your 5 Whys end at something you can actually act on, or at 'people are careless'?
2. Which shortlisted cause does the Lab 7 data actually support?
3. Why must the team not jump straight to solutions here?

Debrief

Bring the class back together and draw out the learning:

- Ask two or three groups to show what they produced, not just what they concluded.
- Ask what surprised them — the gap between what they assumed and what the data showed.
- Link the activity explicitly back to the ANALYZE phase of DMAIC.
- Remind learners this activity rehearses what the Case Study assessment will require.

Reflection questions:

- Think of a recent fix at work — was it aimed at a cause or a symptom?
- What evidence would have proved it either way?

Facilitator notes — what to watch for

- Chains that stop at 'human error' are unfinished — push for the system that allowed it.
- The evidence file deliberately contains both supporting AND confounding items.
- Do not let the group vote before generating — evaluation kills brainstorming.

Success criterion

Each shortlisted root cause is supported by named evidence, and your 5 Whys chain ends at something you can act on.

Assessment link

This activity builds the ANALYZE phase of DMAIC. All activities are in-class activities assessed indirectly through the open-book Case Study (CS), which reuses the same techniques on the same Contoso Service Desk scenario and the same figures (400 tickets, yield 76%, DPMO 50,000). Knowledge is assessed in the Written Assessment (WA-SAQ). Nothing is assessed that is not practised here or taught in the slides.